① Regression

$x^1 \quad y^1 \quad x^2 \quad y^2 \quad - - - \quad -$

feature

$$X \begin{bmatrix} 1 & x^1 \\ 1 & x^2 \\ \vdots & x^3 \\ 1 & x^n \end{bmatrix}$$

$\hat{y}^1$ -> expected

y -> actual

$$\begin{bmatrix} 1 & x_1^1 & x_2^1 \\ 1 & x_1^2 & x_2^2 \\ 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

-> feature 1

-> feat. 2

$$y = w_1 x + w_0$$

Linear regression: $w^* = (X^T X)^{-1} X^T y$

$$X \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \quad y \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

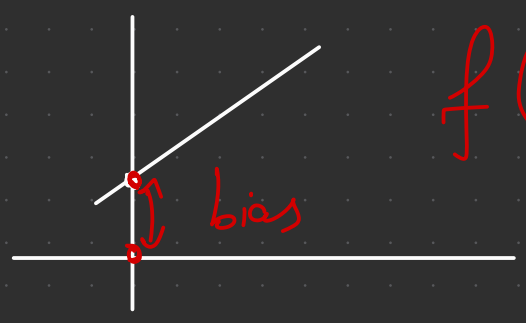
$$w^* = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$$

$$\hat{y} = X w^*$$

$$w \begin{bmatrix} -\frac{1}{3} \\ \frac{1}{2} \end{bmatrix}$$

$$\hat{y} = X \cdot w^* = \begin{bmatrix} \frac{1}{6} \\ \frac{2}{3} \\ \frac{7}{6} \end{bmatrix} \begin{matrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \end{matrix}$$

Ridge Reg. $w^* = (X^T X + \lambda I)^{-1} X^T y$



$f(0) = \text{bias}$

$ax + b$

slope / panta

bias

Elastic

I Sum of errors

$$\sum \left(\hat{y}(i) - y(i)\right) = \left(w_1 \cdot \overset{x_1}{1} + w_0 - \overset{y_1}{1}\right) +$$

$$\left(w_1 \cdot \overset{x_2}{2} + w_0 - (\overset{y_2}{-1})\right) +$$

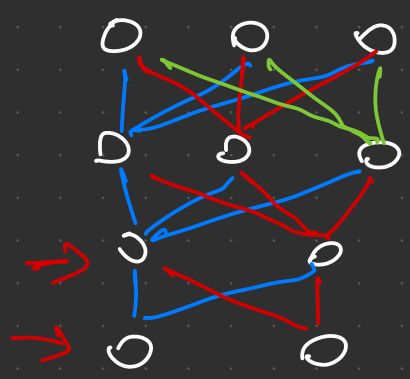
$$\left(w_1 \cdot \overset{x_3}{3} + w_0 - \overset{y_3}{2}\right)$$

$$= 6 w_1$$

6 MLP

My Little Pony

Multi Layer Perception



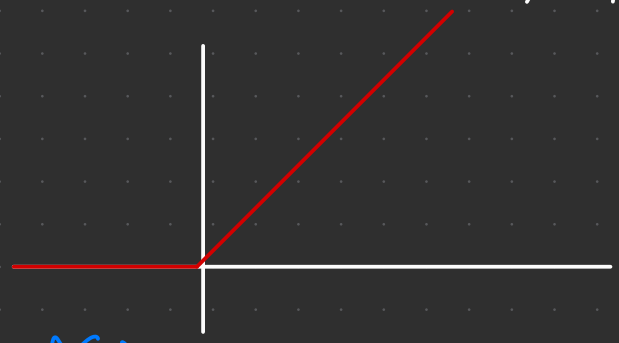
→ output

→ hidden 2

→ hidden 1

→ input

Rectified Linear Unit



$$\mathrm{Relu}(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$\max(x, 0)$

$$x = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$$

$x \in \mathbb{R}^{2 \times 2}$

$n \quad d$

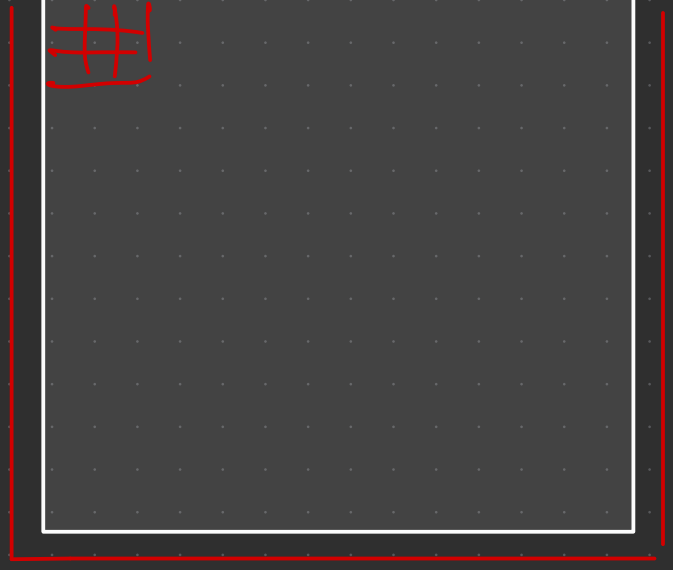
$$\begin{cases} H_1 = \sigma\left(x \underbrace{W^{(1)}}_{2\times 2} + \underbrace{b^{(1)}}_{1\times 2}\right) \\ H_2 = \sigma\left(H_1 \underbrace{W^{(2)}}_{2\times 3} + \underbrace{b^{(2)}}_{1\times 3}\right) \\ O = H_2 \underbrace{W^{(3)}}_{3\times 3} + \underbrace{b^{(3)}}_{1\times 3} \end{cases}$$

RANDOM WEIGHTS

$n \times g \qquad 2 \times 3$

Generate weights, compute Output

$2\times 2 + 1\times 2 + 2\times 3 + 1\times 3 + 3\times 3 + 1\times 3 = 27$



Image

kernel 3x3

N channels

k x k kernel

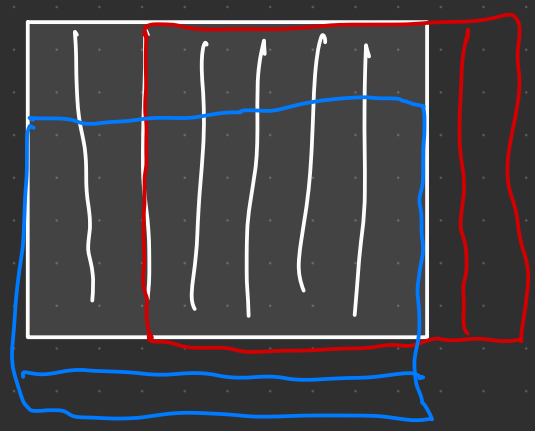
P Padding

S stride

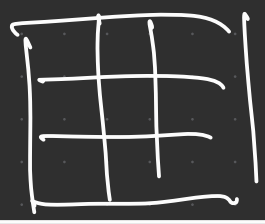
CONV N k P S

2S → kernel 2S

→ stride 2x2



FC S



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input → conv 16 → pool → conv 32 → pool → conv 64 → FC

3ch 16ch / 32ch / 64ch O

Convolution

$$out_h = \left\lfloor \frac{in_h - k_h + 2P_h + s_h}{s_h} \right\rfloor$$

$$out_w = \left\lfloor \frac{in_w - k_w + 2P_w + s_w}{s_w} \right\rfloor$$

Pooling

$$out_h = \left\lfloor \frac{in_h - k_h + s_h}{s_h} \right\rfloor$$

$$out_w = \text{—}\,''\,\text{—}$$

h	w	ch		ch	h	p	s
512	512	3	conv	16	7	3	2

$$out_h = \left\lfloor \frac{512 - 7 + 6 + 2}{2} \right\rfloor = 256$$

$256 \times 256 \times 16$

$FC = \underline{21 \times 21 \times 64 \times 5} + 5$

parameter count conv $= k_h \times k_w \times ch_{in} \times ch_{out}$

bias $= \underline{C_{out}}$

9 Attention

$k_1, v_1 \quad [2,0] \; [-1,1,0]$

$q \quad 1 \quad 2$

$k \quad 2,0 \quad 0,1 \quad \} d$

$$\alpha(q, k_i) = \frac{\exp(a(q, k_i))}{\sum \exp \ldots}$$

$$a(q, k_1) = q^T k_1 \sqrt{\underbrace{d}_{2}} = [1 \;\; 2] \begin{bmatrix} 2 \\ 0 \end{bmatrix} \sqrt{2}$$

$$a(q, k_2) = q^T k_2 \sqrt{d} = [1 \;\; 2] \begin{bmatrix} 0 \\ 1 \end{bmatrix} \sqrt{2}$$

$a(q, k_3) = \ldots$

$a(q, k_4) = \ldots$

$$\alpha(q, k_1) = \frac{\exp(a(q, k_1))}{\exp(a(q, k_1) + \exp(a(q, k_2) \ldots}$$

$$f(q, (k_1, v_1) \ldots (k_4, v_4)) = \alpha(q, k_1) \cdot v_1 + \alpha(q, k_2) \cdot v_2 +$$

(8) RNN 2 hidden

examples = n 2 hidden size = h 4

inputs = d 3 output = q 2

$X \in \mathbb{R}^{n \times d}$ $O \in \mathbb{R}^{n \times q}$

2×3 2×2

$$H_1^1 = \phi\left(\underbrace{X_1}_{2\times3} \underbrace{W_{xh}^{(1)}}_{3\times4} + \underbrace{H_0^{(1)}}_{2\times4} \underbrace{W_{hh}^{(1)}}_{4\times4} + \underbrace{b_h^{(1)}}_{1\times4}\right) \rightarrow \text{step 1 hid 1}$$

$$H_1^2 = \phi\left(H_1^{(1)} W_{xh}^{(2)} + H_0^{(2)} W_{hh}^{(2)} + b_h^{(2)}\right) \rightarrow 2\times4 \text{ hid}_2 \text{ step 1}$$

$$O_t = \underbrace{H_1^{(2)}}_{2\times4} \underbrace{W_{hq}}_{4\times2} + \underbrace{b_q}_{1\times2} \quad \text{out}_1$$

2×2

Num. Param. $\rightarrow$ $W_{xh}^{(1)}$, $W_{hh}^{(1)}$, $b_h^{(1)}$ $\rightarrow$ hidden 1

$W_{xh}^{(2)}$, $W_{hh}^{(2)}$, $b_h^{(2)}$ $\rightarrow$ hidden 2

W_{hq}, b_q $\rightarrow$ output

3×4 4×4 1×4

$4\times4 = 4$ 4×4 1×4

$(2\times3) \times (3\times4) = (2\times4)$

$$\underbrace{d \times h}_{w_{xh}^1} + \underbrace{h \times h}_{w_{hh}^1} + \underbrace{h}_{b_h^1} + \underbrace{h \times h}_{w_{xh}^2} + \underbrace{h \times h}_{w_{hh}^2} + \underbrace{h}_{b_h^2} + \underbrace{h \times g}_{w_{hg}} + \underbrace{g}_{b_g}$$

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② SVM

two-class classification 2D

x_{11} x_{12} x_{21} x_{22}

$(1,1)$ $(2,2)$ $(3,1)$ $+1$ label

$(4,5)$ $(5,7)$ $(6,5)$ $(7,7)$ -1 label

X y

support vector points

hard margin SVM

subject to $y^i\left(w_1 \cdot x_{11}^i + w_2 \cdot x_{12}^i + \overset{\text{bias}}{w_0}\right) \geq 1$

$$\boxed{1\left(w_1 + w_2 + w_0\right) \geq 1}$$

$w_1 + w_2 + w_0 \geq 1$

$2w_1 + 2w_2 + w_0 \geq 1$

$3w_1 + w_2 + w_0 \geq 1$

$-4w_1 - 5w_2 - w_0 \geq 1$

$-5w_1 - 7w_2 - w_0 \geq 1$

$-6w_1 - 5w_2 - w_0 \geq 1$

$-7w_1 - 7w_2 - w_0 \geq 1$

$-w_1 - 4w_2 \geq 21$

3 Trees

$$X \begin{bmatrix} 1 & 3 \\ 3 & 5 \\ 2 & 4 \\ 4 & 7 \end{bmatrix} \qquad Y \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$x_1 \quad x_2$

$x_1 > 1$

$x_1 > 2$

$x_2 > 5$

$x_2 > 6$

Gini / smallest entropy

$$G_h = \sum_{k=1}^{q} \hat{p}_{hk}(1 - \hat{p}_{hk}) = \sum_{k=1}^{q} \hat{p}_{hk} - \sum_{k=1}^{q} \hat{p}_{hk}^2 = 1 - \sum_{k=1}^{q} \hat{p}_{hk}^2,$$

$$H_h = H(\hat{\boldsymbol{p}}_h) = -\sum_{k=1}^{q} \hat{p}_{hk} \log \hat{p}_{hk}.$$

2:2

$x_1 > 2$ $\quad ([1 \;\; 3]^T, 1) \; ([3 \;\; 5]^T, 0)$

$([2 \;\; 4], 1) \; ([4 \;\; 7], 0)$